

POSTER: CLEX: Contract-Bounded Local Execution for Device-Level Network Control

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Abstract

Modern networks increasingly suffer from gray failures that are too fine-grained and short-lived for global re-optimization, while existing local mechanisms lack the context and control boundaries needed for effective bounded response. We propose CLEX, a contract-bounded local execution framework in which the controller defines per-device action boundaries and the device-side execution agent combines local semantics, runtime signals, and short-lived memory to select bounded local policy states that are realized through the local execution substrate. Implementation shows that CLEX enables agile local responses while reducing unnecessary action oscillation.

CCS Concepts

• **Networks** → **Network reliability**.

Keywords

Failure resilience, Network control, Local failure recovery

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1 Introduction

User-visible degradation often first appears as fine-grained gray failures, such as inflated jitter and tail latency on an otherwise reachable path, that require a fast local response [3, 4].

Centralized control provides global visibility but is too heavy for transient local anomalies [5], whereas static local execution is fast but lacks a global action boundary. NetBeacon [9], Brain-on-Switch [8], and RedTE [2] move traffic understanding and fast decisions closer to devices. The remaining difficulty is not traffic understanding alone. Even after classification, the device must select a local action. Common path-centric control mechanisms [7] are policy-driven and less suited to fine-grained local anomalies, where bounded device-local reactions depend on short-lived local history and lifecycle control to avoid overreaction or oscillation.

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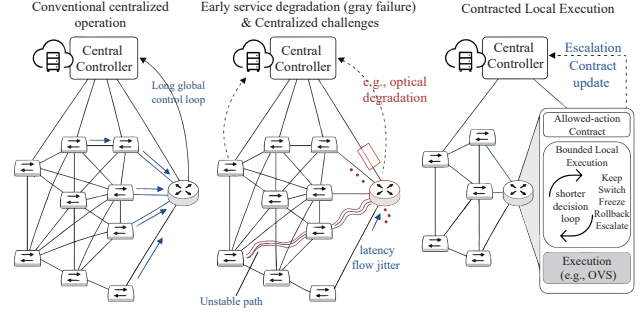


Figure 1: Integrated architecture for gray failure resilience: bounded local response vs. centralized control paths.

CLEX uses semantic signals, runtime hints, and recent local history to select bounded outcomes and manage their lifecycles through hold, freeze, rollback, and escalation. It therefore introduces a contract-bounded device-side execution agent, rather than a static rule engine or packet-driven stateful dataplane. As Fig. 1 shows, the agent enables fast local reactions within controller-defined boundaries to avoid unstable overreaction.

We propose CLEX for turning local anomalies into bounded network actions. It achieves fast device-local reaction through bounded local execution under controller-defined action boundaries. In this way, local reaction can be fast without degenerating into unconstrained online control that rewrites forwarding behavior or triggers unstable network changes, reducing oscillation while preserving deterministic execution.

2 System Design

Figure 2 presents the layered architecture of CLEX. It realizes contract-bounded local execution by separating global admissibility from local execution timing. The controller first derives a per-device admissible action set $\mathcal{U}_i = \{u \in \mathcal{A}_i \mid g_k(i, u) \leq b_k, \forall k\}$, where $\mathcal{A}_i$ contains realizable local actions and $g_k(i, u)$ estimates whether action u remains within service, resource, and forwarding constraints. At runtime, the device-side execution agent operates only inside $\mathcal{U}_i$. Its role is not to synthesize forwarding behavior, but to decide when an admissible outcome should be entered, held, frozen, rolled back, or escalated according to local semantics, runtime state, and short-lived memory. If no admissible local action fits the current condition, the device requests stronger analysis or escalates to the controller.

2.1 Central Contract and Runtime Signals

For node i , the controller encodes the admissible action set $\mathcal{U}_i$, together with trigger conditions and rules into the contract Γ_i . Such actions include switching to a predefined profile, freezing further

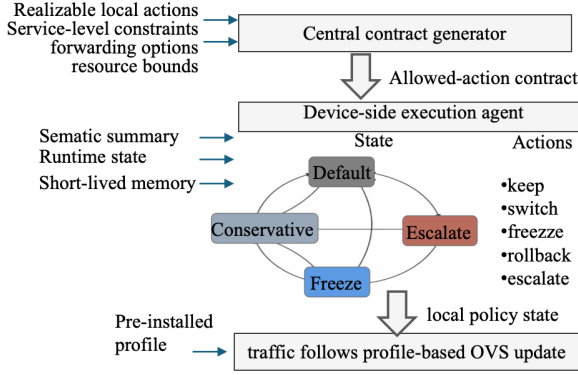


Figure 2: Architecture of CLEX.

local changes, rolling back to the default treatment, or escalating to the controller, rather than arbitrary forwarding-plane rewrites. Contract generation uses the node’s realizable local actions, reachable forwarding options, local resource bounds, and service-level constraints to make local handling useful in as many anomaly situations as possible without violating constraints. At each local decision epoch, the node combines three input. (1) *Semantic summaries from an analysis module*, including the current semantic assessment and its confidence. (2) *Device-local runtime state*, including egress-load summaries and path-health signals. (3) *Short-lived local memory*, including the active profile, hold timer, freeze state, and recent action history.

2.2 Device-side Execution Agent

The device-side execution agent operates under a contract, with actions driven by context and local history rather than static rules. It takes semantic input, local runtime state, and short-lived memory as input, and outputs a bounded local policy state with profile selection, hold/freeze control, so that repeated switching can be suppressed when it may cause oscillation. The agent exposes two interfaces: *request_stronger_analysis(...)* when local confidence is insufficient, and *escalate_to_controller(...)* when the condition is outside the contract boundary or cannot be stabilized locally. This design goes beyond stateful dataplane logic [1], since action selection depends on context, short-lived history, timing, and external outcomes rather than local signals alone. Here, tools denote restricted device-side interfaces exposed under Γ_i . They include sensing interfaces for local observations, analysis interfaces for stronger semantic assessment, and execution interfaces for switching among pre-installed profiles, such as queue selection, meter selection, marking policy, or endpoint redirection.

The agent is triggered by semantic changes or runtime-threshold crossings, rather than by every packet or flow event. Once triggered, it evaluates candidate actions allowed by Γ_i . Thus, the same worsening-jitter signal may keep the default profile under low load, switch to a conservative profile under high load, freeze changes under unstable path state, or escalate after repeated hold periods.

3 System Implementation and Evaluation

We implement CLEX as a lightweight Python daemon on a Linux router with Open vSwitch (OVS) [6]. The controller pre-installs

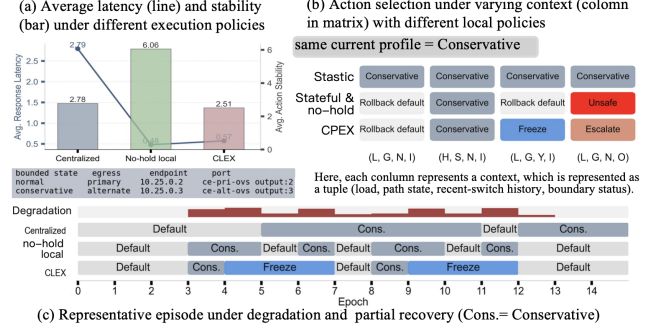


Figure 3: Evaluation of CLEX.

bounded action profiles, and trigger events select the active profile for an affected traffic class; newly arriving traffic follows the update while in-flight flows retain their current handling. We realize queue/profile selection and egress or endpoint redirection through OVS action paths, and implement an automatic degradation/recovery loop in which *freeze* suppresses repeated profile changes and premature rollback.

Fig. 3(a) reports the time to the first effective reaction. CLEX retains the main responsiveness benefit of local handling, staying close to the no-hold local baseline while avoiding the delay of centralized execution. Fig. 3(b) reports execution-state changes under varying context conditions. The no-hold local baseline reacts quickly but switches most aggressively, whereas CLEX suppresses unnecessary state changes without falling back to centralized latency. Fig. 3(c) visualizes a representative episode behind the aggregate results. When the anomaly weakens temporarily, the no-hold local baseline may roll back too early and switch again later, whereas CLEX preserves a bounded local state through the transient interval and avoids unnecessary rollback.

4 Conclusion

We present CLEX for local execution against gray failures. CLEX uses action boundaries defined by controller and device-side bounded execution to achieve early local response without unstable overreaction. The prototype demonstrates that context-driven, low-latency local execution can be realized through profile-based dataplane switching with automatic trigger. Future work includes extending CLEX to multi-node concurrent execution with global consistency guarantees, and refining contract optimization under uncertainty with robustness guarantees for bounded local execution.

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AI disclosure: ChatGPT was used for language polishing and Codex for evaluation-code assistance; authors reviewed all content.

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